

ELFIN CONSULTING

ORGANIZATIONAL LIFE CYCLE ASSESSMENT

Prepared for: ELFIN Consulting, Cologne.
January 2024

PROJECT GENERAL INFORMATION

Project title	The Organizational Life Cycle Assessment of Elfin Consulting, Köln.
Reporting flow	Amount of working hours in 2022
Reporting organization	Entire Elfin Consulting, Köln.
Reference year	2022
Contracting organization	Elfin Consulting, Köln.
Contracting organization website	https://elfin.works/en/home/
Contracting organization main contact	Michael Kühnen, Sustainability Manager (michael.kuehnen@elfin.works)
Report version	First Report (16/March/2023)
	Critical Report Review (Conducting a critical review is under consideration)
	Final Report (26/January/2024)
LCA practitioner	Tiago Furlanetto (tiago.furlanetto@elfin.works)

Table of Contents

PROJECT GENERAL INFORMATION	2
ABBREVIATIONS AND ACRONYMS	4
LIST OF FIGURES.....	4
LIST OF TABLES	4
1. BACKGROUND INFORMATION	5
1.1. Overview of the Organization operation	6
2. GOAL OF THE STUDY.....	7
3. SCOPE OF THE STUDY	8
3.1. Reporting Organization and Reporting Flow.....	8
3.2. System Boundary.....	9
3.3. Time and Geographical Coverage.....	11
3.4. Allocation Procedures	11
3.5. Cut-off Criteria of the Study.....	12
4. LIFE CYCLE INVENTORY.....	12
4.1. Data sources	15
4.2. Software, Environmental Impact Assessment Methods and Indicators.....	16
5. RESULTS.....	18
5.1. Overall Results	18
5.2. Water Consumption impact – contribution analysis.....	20
5.3. Climate change impact – contribution analysis	20
5.3.1. Zoom on the production stage.....	21
5.3.2. Zoom on the End-of-Life stage.	22
5.4. Scenarios Analysis.	22
5.4.1. Scenarios design - alternatives and assumptions.....	22
5.4.2. Scenarios analysis results.....	23
5.5 Sensitivity Analysis.....	25
6. DATA QUALITY ASSESSMENT	25
7. ASSUMPTIONS AND LIMITATIONS	29
7.1. The process for gathering information on capital goods.....	29
7.2. Extrapolating numbers from the survey and inventory	30
7.3. Environmentally extended input-output database	30
7.4. Life Cycle Assessment tools and methods.....	31
8. CONCLUSIONS.....	31
9. REFERENCE:	33

ABBREVIATIONS AND ACRONYMS

CO ₂	Carbon Dioxide
DALY	Disability Adjusted Life Years
EoL	End of Life
eq	equivalents
GHG	Greenhouse gas
IPCC	Intergovernmental Panel on Climate Change
ISO	International Organization for Standardization
LCA	Life Cycle Assessment
LCI	Life Cycle Inventory
LCIA	Life Cycle Impact Assessment
OLCA	Organizational Life Cycle Assessment
PDF*m ² *y	Potentially Disappeared Fraction per Square Meter of land per Year
SA	Sensitivity Analysis
USEEIO	United States Environmentally-Extended Input-Output

LIST OF FIGURES

Figure 1: Life cycle flow diagram for the system analysed.....	7
Figure 2: Life cycle of the services evaluated in this study	10
Figure 3: Flowchart depicting the data, data sources and methods employed in this study.	16
Figure 4: Relative impact attributed to one billed hour (A) and one employee (B).....	19
Figure 5: Contribution of the production-stage process to the water consumption impact.....	20
Figure 6: Contribution of the production-stage process to the climate change impact.....	21
Figure 7: Contribution of the EoL-stage process to the climate change impact.	22
Figure 8: Production stage-related footprints of compared scenarios.....	24

LIST OF TABLES

Table 1 - Impact Categories included in ReCiPe 2016 and used in this analysis.....	13
Table 2 - Impact Categories included in ReCiPe 2016 and used in this analysis.....	17
Table 3: Environmental performance of the reporting organization in each impact category.....	19
Table 4 - Pedigree methodology used to assess data quality.	26
Table 5: Inventory data quality assessment.....	27

1. BACKGROUND INFORMATION

Amidst the evolving global landscape, the urgency for sustainable systems has reached a pivotal juncture. As environmental and societal challenges intensify, there is a critical need for a paradigm shift in our approach to production and consumption. The pressing need for sustainability stems from the recognition that our current practices have far-reaching implications, impacting ecosystems, communities, and the overall well-being of the planet.

In this context, there is a growing awareness of the finite nature of resources and the compelling need to address climate change, biodiversity loss, and social inequalities. The imperative for sustainability is not merely a trend but a strategic necessity driven by the understanding that our actions today profoundly shape the world for future generations.

By embracing sustainability, businesses not only contribute to the preservation of the environment but also align themselves with values that resonate with an increasingly conscious consumer base. It is a call to action that goes beyond compliance, signaling a commitment to resilience, innovation, and positive societal impact. In navigating this compelling need for sustainable business operations, organizations have the opportunity to not only mitigate risks but also thrive in a future where environmental and social responsibility are integral to success.

To comprehensively assess impacts and identify avenues for enhancing the environmental performance of products and services, the Life Cycle Assessment (LCA) emerges as a paramount methodology. LCA, a consolidated tool, serves as the cornerstone for computing and assessing the potential impacts on the environment, human health, and resource depletion throughout every stage of a product or service's life cycle. This powerful instrument transcends mere measurement, actively empowering organizations to strategically enhance environmental performance. LCA is guiding industries toward a future where products and services not only meet but exceed environmental standards, marking a commitment to responsible and environmentally conscious business practices through innovation and optimization.

This study is situated within this broader sustainability landscape. Leveraging LCA, Elfin Consulting has enthusiastically embarked on a proactive exploration, dedicated to more than merely assessing environmental aspects, its commitment extends to enhancing the environmental performance linked to its organizational operations.

By deriving insights from LCA, a nuanced perspective on operational intricacies emerges. In this way, Elfin Consulting lays the foundation of an unwavering commitment to implementing measures that prevent and mitigate potential impacts by identifying opportunities for positive change and shaping practices proactively.

Thus, this study adheres to the principles outlined in the Organizational Life Cycle Assessment (OLCA) definition set forth by ISO/TS 14072. It entails a comprehensive compilation and evaluation of inputs, outputs, and potential environmental impacts at each stage of the organizational operation, adopting a life cycle perspective. Furthermore, it is imperative to note that this LCA aligns with the stringent standards delineated in ISO 14040 and 14044 (ISO 2006a; ISO 2006b).

The following report not only elucidates the methodology, assumptions, limitations, and data sources instrumental in evaluating the potential environmental impacts associated with Elfin's operation in 2022 but also delves into a series of scenarios exploring variations in commute and energy sources.

Crucially, it is underscored that while the intention is to make the study publicly available, our objective is not to assert any comparative conclusions.

1.1. Overview of the Organization operation

Elfin Consulting is headquartered in Cologne, Germany, and has been in business for over ten years. Its personnel is composed of 44 employees.

Elfin Consulting distinguishes itself as a leading management consultancy, notable for its extensive expertise in delivering sophisticated solutions across five pivotal domains: Marketing and communication; Product stewardship and responsibility; Sustainability strategy and implementation; Stakeholder partnerships and engagement; Agile facilitation and digital transformation.

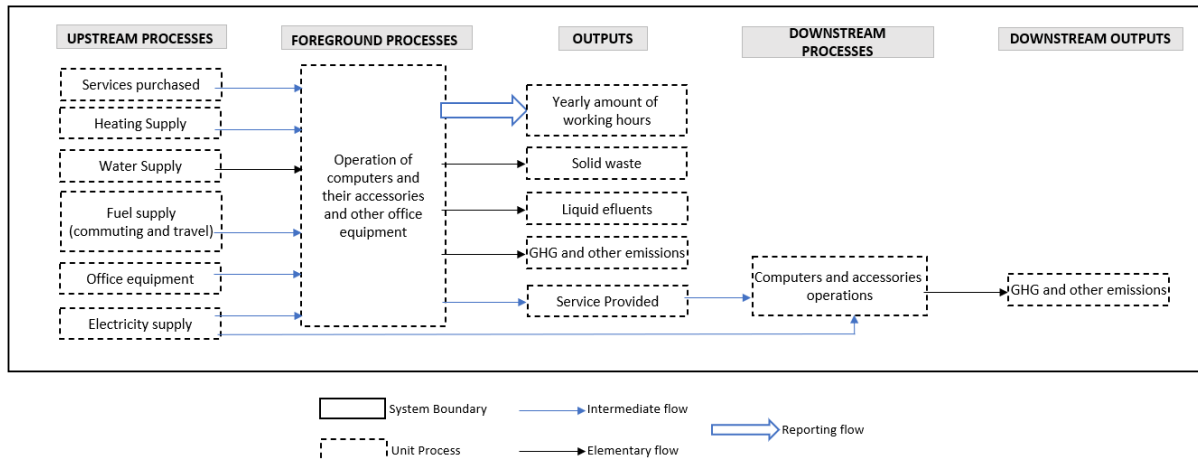
In serving a diverse clientele, the consultancy strategically focuses primarily on industries such as agriculture, crop protection, food production, biotechnology, plastics, and pharmaceuticals. As a provider of advanced knowledge and strategic counsel, the service portfolio encompasses a spectrum of impactful deliverables. These include high-level strategic advisory, professionally conducted seminars, data-driven webinars, immersive workshops, knowledge-rich lectures, and meticulous maintenance of digital platforms.

Capitalizing on a variety of digital file formats, Elfin Consultancy not only shares analysis results but also reflects the extensive service offerings characteristic of respected industry leaders. Going beyond conventional consultancy, Elfin engages in active collaboration with clients to develop and implement strategic decisions that enhance their environmental, social, and governance performance.

In providing invaluable support and guidance to clients, the primary direct activity involves the utilization of electronic devices like laptops, monitors, and smartphones. It is evident that to enable this, other indirect upstream and downstream activities are required, including the procurement of energy and other essential services.

To identify the most critical considerations for Elfin, an exhaustive review of specialized literature was conducted, coupled with talks with internal experts. The pertinent environmental aspects identified through this comprehensive analysis are outlined in Figure 1. This visualization encapsulates all processes and the intricate flow of energy and materials associated with the entire operational framework of Elfin Consulting.

Figure 1: Life cycle flow diagram for the system analysed



It is pertinent to highlight that the reporting flow has the year 2022 as a reference period. Throughout this year, due to the restrictions imposed by the COVID-19 pandemic, work activities were conducted in a hybrid work model. This working approach granted employees the flexibility to seamlessly work remotely, occasionally attending the office.

2. GOAL OF THE STUDY

The present study aims to deliver life-cycle-based insights into potential environmental impacts linked to the operations of Elfin Consulting.

Furthermore, the study scrutinized scenarios with varied assumptions, aiming to pinpoint the conditions under which alternative systems surpass the effectiveness of the current business operation. This rigorous analysis involved meticulously comparing the existing hybrid working model system (as outlined in Table 1 – section 4) with the following range of scenarios:

- (I) **Traditional Model System:** This involves conducting all activities in the office, strategically reducing the frequency of video conferences.
- (II) **Home-Office Model System:** Encompassing the remote execution of all activities, this scenario eliminates the need for commuting, promoting efficiency and flexibility.
- (III) **Four Additional Scenarios with Varying Commuting Frequency:** This set of scenarios systematically explores the impact of commuting, considering frequencies ranging from 1 to 4 days a week. The analyses delve into the nuances of different commuting patterns, providing a comprehensive understanding of how varying levels of on-site presence may influence operational dynamics.
- (IV) **Renewable Energy Integration Scenarios:** A distinctive aspect of these analyses involves the replacement of the current electricity source (national mix) with wind power (the primary source of renewable energy in the region). This deliberate shift toward renewable energy sources underscores the company's commitment to sustainable practices and environmental responsibility, exploring how such an integration can positively impact overall operational efficiency and ecological footprint.

This LCA is strategically employed to establish a foundational benchmark for the organization's environmental footprint. With this baseline, the company can proactively craft strategies aimed at enhancing its environmental performance, providing a tangible means to systematically measure the ongoing progress of its environmental initiatives.

In pursuit of strategic sustainability and marketing goals, this LCA aims to:

- Uncover the organization's environmental impact across LCA impact categories, strategically identifying hotspots to guide future interventions.
- Establish a robust framework for systematically tracking environmental performance, ensuring a comprehensive and dynamic assessment over time.
- Discern the advantages and drawbacks of current working models against potential alternative systems, offering strategic insights for continuous improvement.
- Facilitate internal communication by fostering awareness among employees regarding the environmental impacts of their work. Internal training initiatives will also be implemented to disseminate knowledge of LCA throughout the staff.

Through this LCA report, Elfin aims to showcase its expertise in LCA tools and methodologies, positioning itself as a leading choice for clients seeking insightful environmental analyses.

In essence, this comprehensive LCA report is crafted to deliver meticulous insights into Elfin's environmental performance, catering to both internal stakeholders and external audiences. The strategic integration of scientific rigour and marketing goals ensures that the report informs and clearly demonstrates Elfin's commitment to sustainability and expertise in LCA methodologies.

3. SCOPE OF THE STUDY

The following sections describe the scope of this assessment. It includes a description of the system boundaries, data sources, and methodological framework.

It is noteworthy that the scope definition acknowledged the fact that the present study relates to Organization Life Cycle Assessment. Therefore, following the standard ISO/TS 14072 and instructions given by UNEP (2014), the functional unit or reference flow definition is not used here. Rather two concepts are employed, reporting organization and reporting flow.

Notably, the scope identification underscores the alignment of this study with the principles of Organization Life Cycle Assessment, adhering to the established standards of ISO/TS 14072 and the guidelines set forth by UNEP (2014). This distinctive approach does not employ the traditional use of a functional unit or reference flow definition. Instead, two other concepts, namely reporting organization and reporting flow, must be considered in pursuing a comprehensive assessment.

3.1. Reporting Organization and Reporting Flow

In the realm of product LCA, as elucidated by Weidema (2004), a crucial aspect involves the use of a functional unit (FU) to compare alternatives that serve similar functions for the user or consumer. The FU acts as a standardized measure, ensuring a fair and quantitative comparison of alternatives designed to fulfill the same purpose. In the context of product LCA, the reference flow signifies the outputs (final product/service) from processes within a specified system, essential for accomplishing the function expressed by the functional unit (ISO 2006a). This approach aims to provide a clear and equitable basis for comparing alternatives, meeting the needs of diverse audiences seeking transparency and robust decision-making in their sustainability assessments.

On the other hand, as argued by Martinez-Blanco et al. (2015), establishing a universally shared function between organizations may prove impractical. This challenge arises from the myriad differences that can exist among companies, even those within the same sector. Such differences encompass unique product portfolios and distinct business models, making direct functional comparisons challenging. Consequently, the authors assert that comparisons between organizations lack a consistent pattern, highlighting the effectiveness of Organizational Life Cycle Assessment (O-LCA) in diagnosing potential impacts and monitoring performance over time. Furthermore, in line with UNEP/SETAC (2015), the concept of reporting flow is introduced, representing the measure of outputs from the reporting organization during the reference period

Hence, within an O-LCA context, the "reporting organization", which represents the organization under study, replaces the "functional unit" as the unit of analysis. While "reporting flow" substitutes the "reference flow" as the measure of the outputs.

In this study, the analytical focus extends to the entire reporting organization, comprising all five departments and a total of 44 employees. The consolidation of the entire company as a singular unit of analysis is made feasible by Elfin's compact size and localized operations. This strategic decision is further substantiated by the uniform legal, financial, and organizational structures across its business operation areas. Additionally, each of the five areas of expertise within Elfin presents a comparable portfolio of services, showcasing similar value chains and environmental interventions. This intentional alignment amplifies the coherence and robustness of the assessment.

To complete and base inventory of this study, establishing a quantitative linkage of Elfin's value chain with its service portfolio output, the present study adopted the "total billed hours in 2022" as the reporting flow.

The selection of this unit of measurement is in harmony with that recommended by UNEP/SETAC (2015) and took into account the equivalent nature, annual sales, and environmental profile of the services, ensuring, in this manner, a robust representativeness of the portfolio and outputs of the reporting organization.

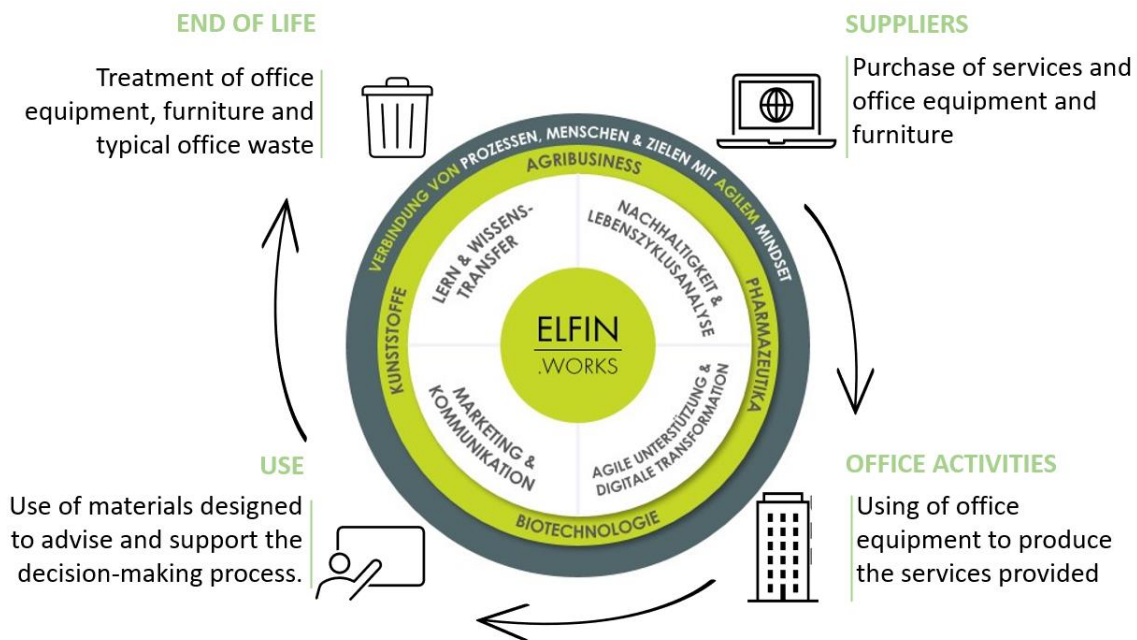
In order to accomplish a thorough inventory for this study and establish a quantifiable connection between Elfin's value chain and its service portfolio output, the chosen metric to represent the reporting flow is the "total billed hours in 2022," which aligns with the recommendation by UNEP/SETAC (2015). This selection takes into careful consideration the equivalent nature, annual sales, and environmental profile of the services. By doing so, the study ensures a meticulous representation of the reporting organization's portfolio and outputs, emphasizing precision and transparency in its environmental assessment.

3.2. System Boundary

The methodology implemented in this study encompasses the activities within the reporting organization and other elements in the value chain that are indirectly associated with Elfin's service provision. To ensure a thorough understanding of its impacts, this assessment meticulously incorporates all relevant and traceable upstream, foreground, and downstream processes. For instance, when assessing the environmental aspects of company-owned equipment, the study extends beyond mere energy consumption. It delves into the impacts associated with additional processes and inputs essential for both the equipment and energy production. This comprehensive approach provides a holistic perspective, offering a nuanced insight into the environmental footprint associated with Elfin's services while reinforcing a commitment to rigorous analysis in sustainability assessments.

At this point, it is worth mentioning the absence of indirect downstream activities linked to the distribution of services. Yet, given the intangible nature of the service (non-physical product), processes associated with the end of life (EoL) were not included. However, environmental considerations pertaining to the EoL of office equipment, furniture, and the management of standard office waste have been thoughtfully incorporated into the analysis. Figure 2 illustrates the life cycle under analysis.

Figure 2: Life cycle of the services evaluated in this study



Throughout the scope definition process, integrating the use phase and articulating a representative product delivered by the reporting organization posed a considerable challenge. This challenge arises from Elfin's distinctive offering of intangible knowledge as a service, rather than tangible goods.

Nevertheless, to comprehend how clients engage with the documents provided by Elfin and to estimate the potential environmental impacts stemming from this interaction, a pragmatic definition of a "typical product" and its utilization by clients was derived. This definition was built by analysing a set of representative files, comprising only the final product delivered to the client and excluding preliminary versions and support materials. The formulation of this definition involved collaboration with all leaders and the managing director of the reporting company.

Based on the assumption, a typical Elfin product takes the form of a digital file, such as a PDF or PowerPoint document, with an approximate size of 5.13 MB. The envisaged usage pattern is as follows:

- Following Elfin's internal development of the product, the file is initially transmitted to the client for an initial review.
- During the first client review, an average of 30 minutes is spent by clients to assess the document and propose enhancements.
- Upon completion of the first review, the file is returned to Elfin. Following the implementation of requested refinements, the revised file is sent back to the client.

- Subsequently, the client dedicates another 30 minutes to review and approve/accept the document.
- Upon the project's conclusion, the client utilizes the file for a single presentation to an audience (colleagues/supervisors/stakeholders). This presentation of slides crafted by the client is expected to last approximately 20 minutes.
- Ultimately, the document is archived on the client's data hosting service.

In relation to the review process, the assumption is made that this activity necessitates the operation of a single laptop by the client. Concerning the presentation, it is presumed that the host person (i.e., the client) shares the information with five individuals via videoconference. Consequently, this activity involves the operation of six laptops.

Furthermore, the estimation of the total quantity of data stored by all clients utilized the representative set of files. Taking into account the average size of a representative file, the average number of billable hours specifically dedicated to its preparation, and the total billable hours of the organization within the reference period, it became feasible to calculate the overall volume of stored data.

3.3. Time and Geographical Coverage

The data and assumptions considered in this study aim to accurately depict the equipment, processes, and market conditions relevant to Elfin's organizational operations in 2022. As such, the reference period for the primary data collected is specifically 2022. Furthermore, the data is site-specific, offering a detailed insight into the activities conducted within the reporting organization's boundary.

Secondary data were employed when direct information collection was impractical, primarily due to time and availability constraints. Regional or national industrial statistics were utilized to bridge this gap. Typically, this statistical information spans the last seven years (2015-2022). Yet, the background processes were taken from the most recent version ofecoinvent (3.9).

In this study, Germany is selected as the geographical reference unit due to the organization's headquarters in Cologne, its employees residing in North Rhine-Westphalia, and the primary execution of activities within German territory. However, in instances where ecoinvent lacked specific information about Germany, data related to Europe was utilized. When neither of these options was available, data representing the global context was incorporated.

3.4. Allocation Procedures

Following the guidelines outlined in ISO 14044, when the system in focus generates multiple products with economic value (co-products), an LCA practitioner, aiming to discern the impacts associated with each product individually, should either expand the system boundaries to include adjacent systems or allocate the impacts of co-products based on coherent physical or economic relationships.

As there are no co-products within the system under analysis in this study, none of these procedures was applied. Despite the reporting organization having five areas of activity, there is a standard product common to all areas, as interpreted according to the description in section 3.2.

The only exercise done here was related to capital goods (electronic devices and furniture, for instance) and the fact that they are used over a longer time than the reference period (one year). In this manner, to determine the fraction of the total impact of a good's production assigned to one year of use, a division method was employed, taking into account the life expectancy of each capital item.

It is essential to note that the background processes provided by ecoinvent, utilized in this study, adhere to the "cut-off by classification" model as the standard. This model is practical for production systems dealing with materials, resources, or energy from upstream or downstream processes, describing environmentally relevant physical flows between interconnected processes. However, it does not account for the benefits of alternative End-of-Life (EoL) treatment, following an attributional allocation approach.

3.5. Cut-off Criteria of the Study

This section elucidates the criteria applied in the current analysis, focusing on the delineation of the system's boundary. It's imperative to distinguish this from the cut-off by classification method utilized in the ecoinvent database.

As explicated in section 3.2, detailing the approach for defining the system's boundary to align with the study's objectives, no specific cut-off point has been designated for this analysis. As previously stated, all data pertaining to energy and material flows from relevant and traceable upstream, foreground, and downstream processes have been encompassed in the model.

In instances of uncertainty, conservative assumptions were employed. Additionally, no proxy data were utilized, as the databases provided comprehensive information on all flows and processes within the system.

4. LIFE CYCLE INVENTORY

There is a direct and strong correlation between the quality of the data used and the quality of the LCA results. In this study, diligent efforts were undertaken to acquire the most accurate, comprehensive, consistent, and representative data within the confines of the study's goal and scope, while mindful of time constraints.

Table 1 presents the inventory compiled and used to compute the company's impacts.

Table 1 - Impact Categories included in ReCiPe 2016 and used in this analysis

SOURCE: RIVM Report 2016-0104.

ACTIVITY	OBJECT	VALUE	UNIT	SOURCE
Home-office energy consumption	Total consumption of electricity from local mix grid	702.1	kWh	Employee survey
	Total consumption of heating from local supply	207.5	GJ	
Office energy consumption	Total consumption of electricity from local mix grid	68.04	kWh	Inventory
	Total consumption of heating from local supply	2.15	GJ	Inventory
	Total electricity consumption (Office equipment)	13458.10	kWh	Inventory
Water	Total water usage	362.63	m ³	Shariff, 2009
Employee commutation	Total distance by own car/taxi	480	km*person	Employee survey
	Total distance by train/subway	12912	km*person	
	Total distance by bus	–	km*person	
	Total distance by own e-bicycle	–	km*person	
	Total distance by bicycle	291,6	km*person	
	Total distance walking	109.2	km*person	
Business travel	Total distance by plane	–	km*person	Employee survey
	Total distance by car	3984	km*person	
	Total distance by train	4110	km*person	
	Total distance by bus	15	km*person	
Office equipment and furniture	Laptop	71	unit	Inventory
	Monitor	45	unit	
	Mouse	45	unit	
	Keyboard	45	unit	
	Mobile Phone	44	unit	
	Office chair	45	unit	

	Desk	45 unit	
Waste generated through office	Total commercial waste, mix	7,2 ton	destatis.de/
	Total waste water	108.8 m3	Schleich & Hillenbrand, 2009
Purchased Service	Elfin's own marketing and advertising	– Euros	Inventory
	Operations and Maintenance	– Euros	
	Banking services	– Euros	
	Postal service	– Euros	
	Hotel services	– Euros	
	Legal services	– Euros	
	Real state service	– Euros	
	Data processing, hosting, and related services	11760.8 Euros	
	Accounting, tax preparation, bookkeeping, and payroll services	– Euros	
	Commissioned companies	– Euros	
Other Services	internet access (work, 0.2 Mbit/s)	56111 hours	Inventory and Survey
	internet access (video-conference, 0.7 Mbit/s)	27695.5 Euros	
	Total time of clients laptop operation (review)	52.98 hours	Inventory
Use Phase of sold service	Total time of clients laptop operation (presentation)	108.09 hours	
	Total cost to store files on clients data hosting service	0.0063 USD	

4.1. Data sources

The primary data were directly compiled from the reporting organization and complemented with information from published sources.

Most of the information related to the foreground processes was acquired through an inventory of the organization's capital goods and a survey conducted with the organization's employees. Upon receiving the information, a thorough cross-check was conducted to ensure completeness and consistency, considering the company's activities, the number of employees, and the insights of the company's management. The data provided by the company and its employees included capital goods, electricity and heating consumption, business travel, commuting, virtual meetings, and expenses associated with services.

Whenever available, data directly provided by manufacturers were utilized to calculate impacts associated with the use phase of capital goods (e.g., the energy consumption of electronic devices). Despite efforts to contact manufacturers, accessing data on producing goods was not possible. Consequently, the study relied on average data to estimate their production and end-of-life (EoL) impacts.

Regarding energy consumption, the survey allowed for the estimation of the area in employees' houses exclusively dedicated to work-related activities. Secondary data indicative of national household energy consumption for space heating were used to calculate heating energy consumption. Electricity consumption was calculated considering the power of the lamps used in the working space. As mentioned earlier, manufacturers' information was used to estimate the electricity consumption of electronic devices.

Moreover, some environmental aspects associated with regular office activities were estimated based solely on secondary data. Statistics on water use in office buildings in the European Union served as the basis for calculating the organization's water consumption. An estimate of the wastewater generated by the organization was derived from the national average demand for wastewater treatment. Elfin's office waste was estimated using North Rhine-Westphalia regional figures.

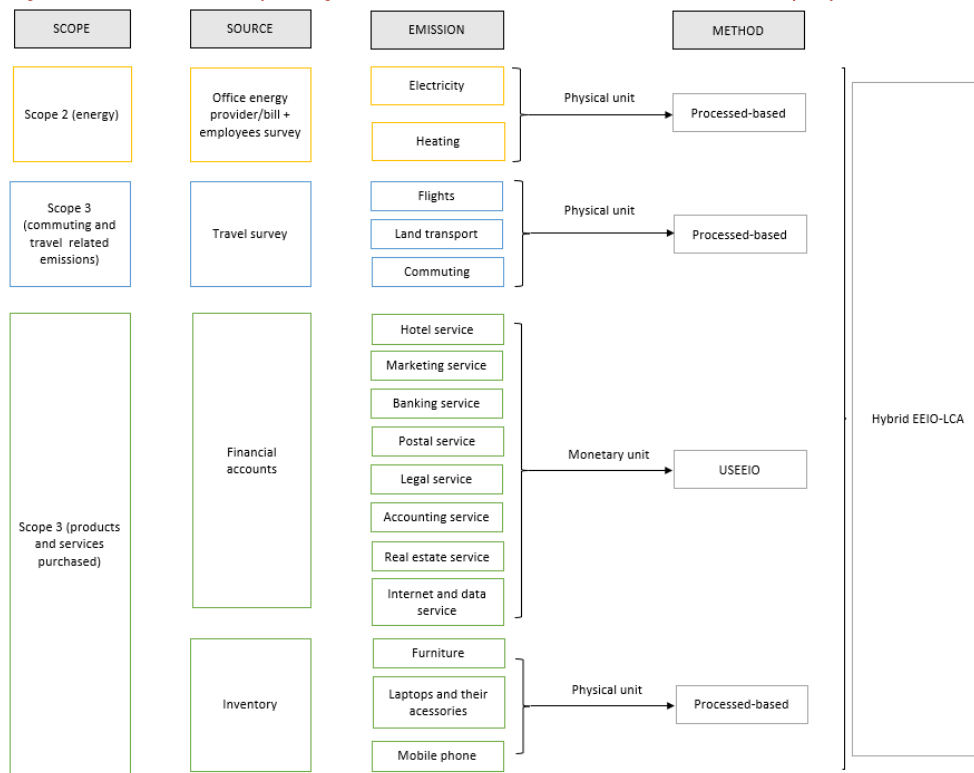
An overview of the data sources associated with the activities performed within the reporting organization is presented in Figure 3, and further details can be found in the Annex Supplementary Information.

Background processes data were mainly obtained from ecoinvent v3.9.1 and, in a few cases, from the US Environmentally-Extended Input-Output (USEEIO) v2.0. These are the two databases integrated into the LCA software used.

Ecoinvent version 3.9.1 was chosen as it is the most up-to-date and comprehensive version of the ecoinvent database, acknowledged as one of the most complete background Life Cycle Inventory (LCI) databases available. Whenever possible, national unit processes were selected to represent local specificities. In the absence of country-specific information, this work prioritized region-specific (EU) and global inventories.

USEEIO v2.0 represents the most recent version of this macroeconomic method that connects sectors of the US economic structure, providing a detailed representation of the financial transactional flows and associated environmental accounts.

Figure 3: Flowchart depicting the data, data sources and methods employed in this study.



4.2. Software, Environmental Impact Assessment Methods and Indicators

The system under analysis was modelled using Earthster (earthster.org), a life cycle software system developed by specialists from Harvard University and collaborators.

To compute the impacts, translating the elementary flows into environmental impacts, this study used the life cycle impact assessment (LCIA) ReCiPe 2016 method. The analysis encompassed two midpoint impact categories (climate change and water use) and the three endpoint areas of protection (damage to ecosystems, human health, and resource availability). However, the results and discussion primarily focus on categories that are particularly relevant to the reporting organization – climate change and water use. Descriptions of impact categories and areas of protection can be found in table 2. For a more detailed and comprehensive description of the ReCiPe 2016 methodology, refer to the following link.

Table 2 - Impact Categories included in ReCiPe 2016 and used in this analysis.

SOURCE: RIVM Report 2016-0104.

Impact Category / Indicator	Unit	Description
Climate change	kg CO ₂ -eq	Indicator of potential global warming due to emissions of greenhouse gases to air. Divided into 3 subcategories based on the emission source: (1) fossil resources, (2) bio-based resources and (3) land use change.
Water use	m ³	Indicator of the relative amount of water used, based on regionalized water scarcity factors.
Damage to Human Health	DALY	Human health measures the impacts of emissions of harmful substances having effects on human health through their toxicity (carcinogenic and non-carcinogenic), their effect on the respiratory system (respiratory inorganic and respiratory organic), their effect on ozone depletion or related to ionizing radiations. This indicator is based on IMPACT 2002+ vQ2.28 (Jolliet et al. 2003; Humbert et al. 2012). It is measured in DALYs (Disability Adjusted Life Year). It excludes the health effects of nutrition and only focuses on indirect effects.
Damage to Ecosystems	PDF-m ² -year	Impact on ecosystem / Ecosystems quality indicator measures the potential impact on ecosystems (biodiversity, species and their inhabitant) caused by emissions of harmful substances associated with a product, process or organization. It takes into account aquatic ecotoxicity, terrestrial ecotoxicity, terrestrial acidification and eutrophication, aquatic eutrophication and aquatic acidification (water turbidity and land use are excluded). It characterizes the fraction of species disappeared on one m ² surface during one year. This indicator is based on IMPACT 2002+ vQ2.21 (Jolliet et al. 2003; Humbert et al. 2012). The impact metric is expressed in PDF-m ² -y ("Potentially Disappeared Fraction of species over one m ² and during one year").
Damage to Resource Availability	USD	It focuses on the future costs for extraction of the resource as a consequence of current depletion, and these divide into costs in the form of energy or exergy use for future extraction. Measured in monetary costs (USD).

It is noteworthy that midpoint impacts are typically based on a relatively robust impact modelling approach. In comparison, endpoint categories are estimated using the best available knowledge, but a

high level of uncertainty is typically associated with the pathways that connect sources of emissions and final endpoint impacts.

5. RESULTS

This chapter unveils the environmental performance results of the reporting organization in 2022, coupled with an in-depth exploration of the various scenarios evaluated in this study. The initial section (5.1) furnishes a comprehensive overview of the results pertaining to the impact categories encompassed in the analysis. Following that, section 5.2 scrutinizes the impacts related to climate change and water use, with the contribution analysis spotlighting the specific hotspots within each process. These two categories were chosen due to their significance to the company, as highlighted earlier, and their calculations involve lesser uncertainty. The complete set of indicators is available in the supplementary information.

5.1. Overall Results

The assessment of the organization's potential impacts during the reporting period revealed the embodiment of 254,000 m³ of water throughout its life cycle activities, accompanied by a carbon footprint of 36 tons of CO₂eq. Additionally, the company can be attributed relative responsibility for the potential impact on 0.003 species annually, the virtual probability of reducing a person's life expectancy by 1.32 years, or diminishing the availability of resources, thereby increasing the cost of providing them by 161,000 USD.

In general, the production stage emerges as the primary hotspot in all impact categories. Water use is the most prominent example, with 99.6% of the impact occurring during the production phase. As indicated in table 3, the production stage, encompassing all activities within the company's boundaries and upstream processes facilitating production activities, accounts for over 97% of four out of the five impact categories. An exception is the climate change impact, where the production stage still constitutes the majority of the environmental burden, contributing approximately 89%, while the End-of-Life (EoL) stage accounts for nearly 11%.

This is a reasonable outcome, given the nature of the company's business model and its services. Typically, most processes in a consultancy firm's value chain occur at the production stage or upstream. The use phase-related impacts are usually negligible, as consultancy firms specialize in providing expertise and advice rather than physical products.

Similarly, the EoL impacts are likely reduced. Several factors contribute to the relatively small impacts associated with the end-of-life phase, including the characteristics of typical office waste (mainly paper, cardboard, and organic waste) and the relatively long lifespan of capital goods. Capital goods usually have a longer life expectancy than the reporting period (one year). Therefore, for a fair assessment, their waste treatment-related impacts are scaled according to the years of useful life, resulting in a reduction of their end-of-life impacts.

Table 3: Environmental performance of the reporting organization in each impact category
 for the year 2022 | t (ton 10³); n (nano 10⁻⁹); μ (micro 10⁻⁶); m (milli 10⁻³).

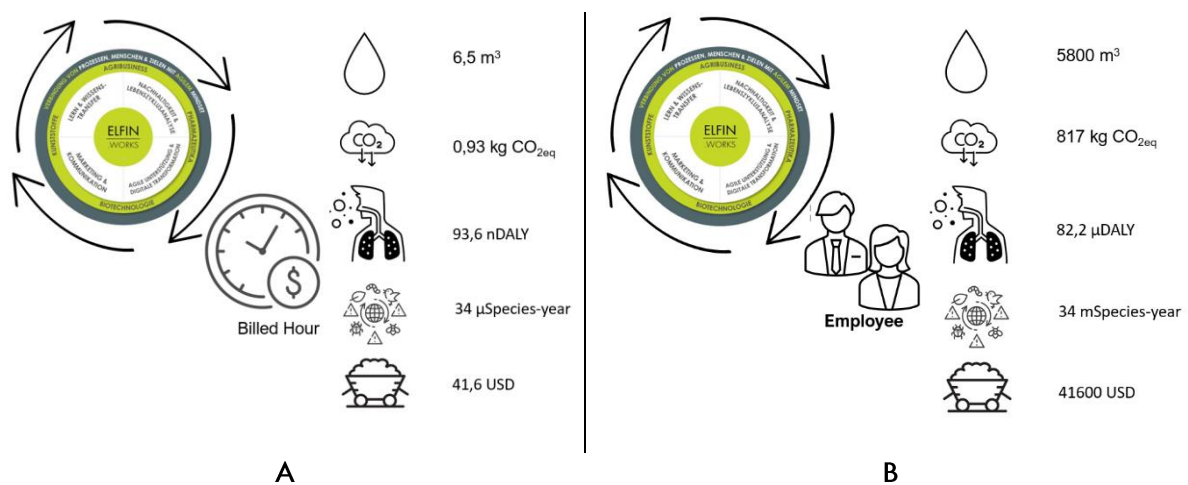
STAGE	Water use (m ³)		Climate-change (T CO ₂ eq)		Damage to ecosystems		Damage to human health		Damage to resource availability	
	1000 m ³	contribution (%)	t CO ₂ eq	contribution (%)	m species-year	contribution (%)	DALY	contribution (%)	1000 USD2013	contribution (%)
PRODUCTION	253	99.60%	32	88.94%	0.004	99.27%	1.29	97.74%	1570	97.85%
USE	0.26	0.10%	0.005	0.01%	0.000003	0.10%	0.001	0.09%	0.26	0.02%
END OF LIFE	0.76	0.30%	4	11.04%	0.00002	0.64%	0.03	2.17%	34.3	2.13%
FULL CYCLE	254	100%	36	100%	0.004	100%	1.32	100%	1610	100%

Following the figure, each billed hour triggered the consumption of 6,3 m³ of water or the emission of 0,93 kg of CO₂ equivalent. The work conducted by one employee potentially impacted the quality of human life by 82,2 micro (10⁻⁹) DALYs, damaged the quality of the ecosystem by 34 milli (10⁻³) species-Year and affected the availability of resources by 41.600 USD.

Figure 4 presents the results across various impact categories, serving as a benchmark for monitoring the organization's environmental performance in the future. The figure depicts the contribution of each billed hour in 2022 or an individual employee to different impact categories.

According to the figure, each billed hour led to the consumption of 6.3 m³ of water or the emission of 0.93 kg of CO₂ equivalent. The work performed by one employee potentially impacted the quality of human life by 82.2 micro (10⁻⁹) DALYs, affected the quality of the ecosystem by 34 milli (10⁻³) species-year, and influenced the availability of resources by 41,600 USD.

Figure 4: Relative impact attributed to one billed hour (A) and one employee (B)
 in 2022. n (nano 10⁻⁹); μ (micro 10⁻⁶); m (milli 10⁻³).



5.2. Water Consumption impact – contribution analysis

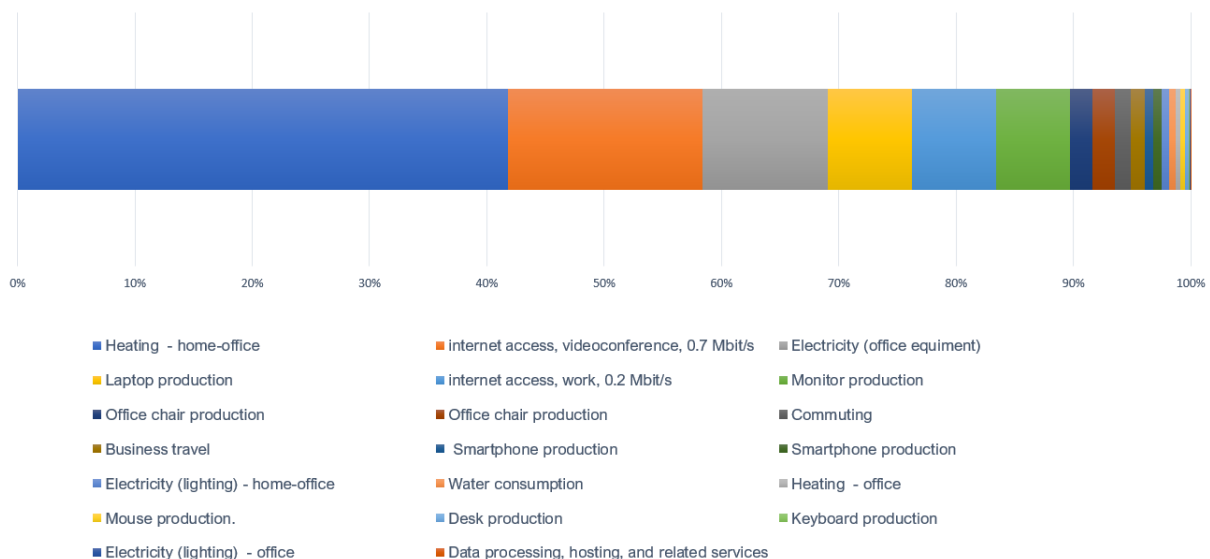
The preceding section unveiled that within the organization's value chain, water consumption is predominantly driven by the production stage, with 99.6% of all water consumption attributed to this phase. Consequently, the contribution analysis will exclusively scrutinize this life cycle phase.

As per Figure 5, the top five processes responsible for propelling water consumption, collectively contributing to over 83% of the water use impact, are: home-office heating (41.8%), internet use for online meetings (16.5%), electricity required for the operation of electronic devices (10.7%), internet use for work (7.2%), and the production of laptops (7.2%).

A unified grouping of all energy consumed accounts for 53.6% of the organization's water footprint. This stems from the inclusion of all water required in the fossil fuel extraction and manufacturing processes within the background inventory. According to Ou et al. (2015), up to 4,833 litres of water are necessary to produce 1 MWh in a coal power plant, and up to 1,590 liters of water are needed for the same energy output from natural gas. These two energy sources constituted nearly 70% of the German electricity mix in recent years (IEA, 2021).

The substantial water consumption related to internet use for work and videoconferencing can be attributed to the significant amount needed to power and cool down data processing centers. Mytton (2021) asserts that in 2018, a single large data center consumed up to 15.8 billion liters of water over the year.

Figure 5: Contribution of the production-stage process to the water consumption impact.



5.3. Climate change impact – contribution analysis

Climate change impact is driven by greenhouse gases (GHG) like carbon dioxide (CO₂), methane (CH₄), dinitrogen monoxide (N₂O), hydrochlorofluorocarbons (HCFCs), hydrofluorocarbons (HFCs) and ozone (O₃) in the lower atmosphere. Emissions of a GHG lead to an increased atmospheric concentration of these gases which, in turn, amplify their radiative forcing capacity (w/m²), eventually, contributing to a rise in the global mean temperature (IPCC, 2022a).

As per the Intergovernmental Panel on Climate Change (IPCC, 2022b), the predominant source of anthropogenic GHG emissions arises from the combustion of fossil fuels for energy production and the manufacturing of fossil-based materials. In the current study, this assertion held true. Considering all life cycle phases, electricity and heating consumption accounted for 52.15% of the total GHG emissions.

A comprehensive contribution analysis of each process to the production and end-of-life stages is provided in the subsequent sections. The use phase was omitted from the analysis, as the emissions at this stage represent less than 0.5% of the total.

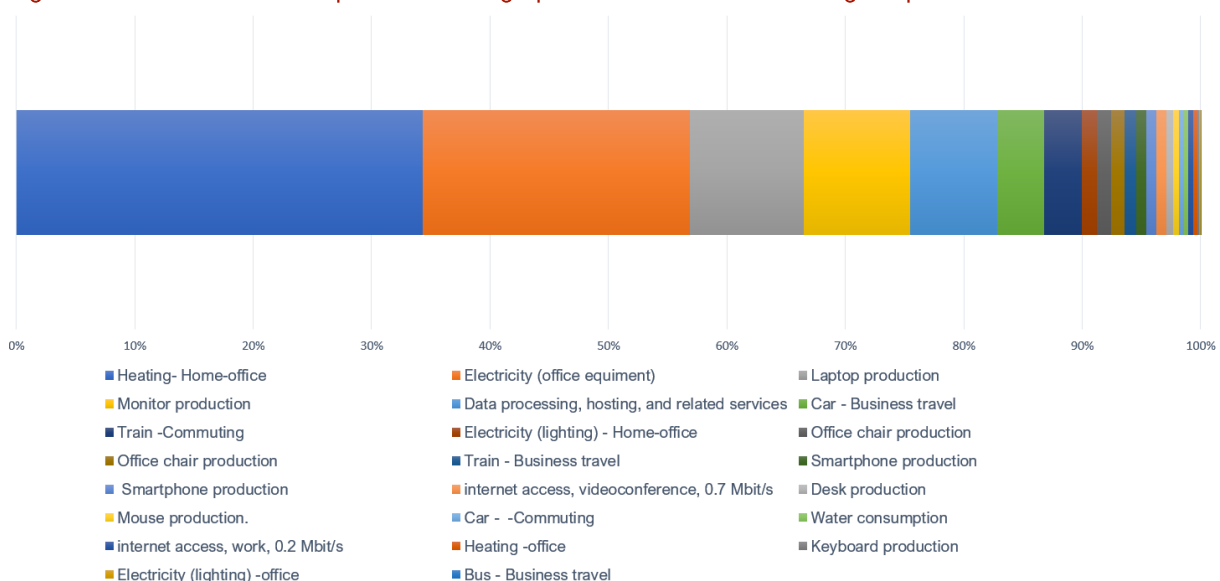
5.3.1. Zoom on the production stage.

In 2022, the production phase was accountable for the emission of 32 tons of greenhouse gases (GHG), representing approximately 89% of the organization's total carbon footprint. As previously elucidated, this significant fraction attributed to the production phase arises from the fact that this stage encompasses the majority of processes and consumes the most resources throughout the life cycle.

Figure 6 illustrates how each process listed in the production phase contributes to the climate change impact. According to the figure, the most substantial contributors to production stage emissions are the consumption of energy, with home-office heating accounting for 34.4% and electricity required for the operation of electronic devices accounting for 22.5% of all stage-related emissions. Other notable emission sources include laptop production (9.7%), monitor production (9%), and data storage on host servers (7.3%). These three processes are also energy-intensive and demand significant material inputs, which may explain their relatively high contribution share.

It is noteworthy to emphasize the comparatively minor contributions of business travel and commuting. Typically, these processes are GHG emission-intensive and constitute a significant source of emissions in organizations with characteristics similar to the one reported here. However, the COVID-19 pandemic imposed sanitary restrictions on employees in 2022, leading to a reduction in employee travels to work and clients.

Figure 6: Contribution of the production-stage process to the climate change impact.



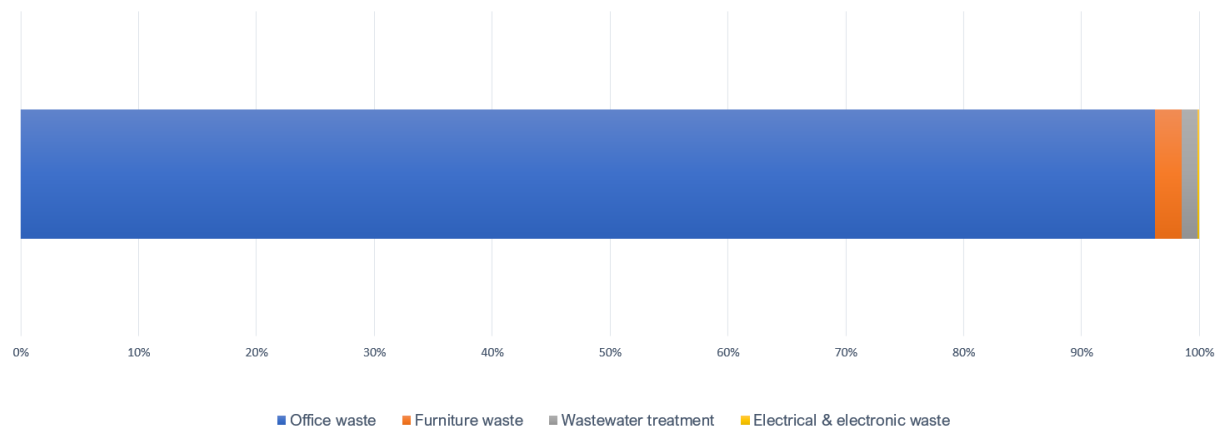
5.3.2. Zoom on the End-of-Life stage.

In 2022, the carbon footprint of the End-of-Life (EoL) phase was 3.97 tons of CO₂eq, constituting approximately 11% of the organization's total emissions.

Figure 7 delineates how each process listed in the EoL phase contributes to the climate change impact. As depicted in Figure 7, GHG emissions associated with the EoL phase are primarily dominated by office waste treatment (96.2%). Office waste, predominantly composed of paper, cardboard, and organic waste, emits greenhouse gases through decomposition or recycling processes. However, rather than emission factors, the significant contribution of office waste treatment to the carbon footprint was determined by the estimated quantity generated - up to 626 times greater than other types of waste.

It is crucial to note that office waste figures were derived from regional statistical data (Genesis, 2022), indicating that each household generates 160 kg/year of domestic waste similar to office waste. While this represents an approximation, it is considered a conservative approach.

Figure 7: Contribution of the EoL-stage process to the climate change impact.



5.4. Scenarios Analysis.

5.4.1. Scenarios design - alternatives and assumptions.

In order to furnish the organization with a robust foundation for devising strategies to enhance its environmental performance, this study conducted an exploration of scenarios to analyse how potential modifications in key processes within the current system could impact the organization's water and carbon footprints.

Considering that production holds the predominant influence on both footprints and encompasses all processes under the organization's control, the scenario analysis exclusively focused on the production phase. It was assumed that other life cycle stages would remain unchanged. Additionally, the following conditions were established: (i) all parameters were adjusted to mirror the conditions of full-time employment for all 44 employees, and (ii) each scenario was evaluated based on the same probability of occurrence.

Following the end of the pandemic and the easing of sanitary restrictions, organization employees are now able to attend the office on a daily basis. Consequently, it was deemed reasonable and feasible to investigate how an increased frequency of commuting could impact the organization's environmental performance.

The escalation in office presence, aside from influencing the intensity of transport activities, has implications for other processes, including energy consumption at home and in the office, as well as internet use in videoconferences, given that meetings between company employees could occur on the office premises.

Considering these factors, six scenarios were devised for comparison with the current baseline framework (scenario 0):

- Scenario 1 - All employees work exclusively from home. Therefore, there is no commuting.
- Scenario 2: All employees commute at least once a week.
- Scenario 3: All employees commute at least twice a week.
- Scenario 4: All employees commute at least 3 times a week.
- Scenario 5: All employees commute at least 4 times a week.
- Scenario 6: All employees work exclusively in the office. So, commuting 5 times a week.

Note that the scenarios evaluate changes in commuting patterns, not predicting any increase or decrease in business travel rate.

Additionally, other six scenarios were designed to assess the implications of different electricity sources. Some companies provide wind-powered electricity in the region where the organization is headquartered. Thus, the new set of scenarios incorporates the same characteristics of the previous alternatives, and additionally includes the use of wind-powered electricity.

- Scenario 7: All employees work exclusively from home. Therefore, there is no commuting. The house is entirely powered by renewable electricity sources.
- Scenario 8: All employees commute at least once a week. Office and house are entirely powered by renewable electricity sources.
- Scenario 9: All employees commute at least twice a week. Office and house are entirely powered by renewable electricity sources.
- Scenario 10: All employees commute at least 3 times a week. Office and house are entirely powered by renewable electricity sources.
- Scenario 11: All employees commute at least 4 times a week. Office and house are entirely powered by renewable electricity sources.
- Scenario 12: All employees work exclusively in the office. So, commuting 5 times a week. The office is entirely powered by renewable electricity sources.

5.4.2. Scenarios analysis results.

Figure 8 provides a comprehensive view of the analysis outcomes across all 12 scenarios tailored for the production stage.

In terms of the carbon footprint and resource availability impact, it is evident that increased commuting translates into higher carbon and resource footprints. This is primarily due to the substantial consumption of fossil fuels and the consequent greenhouse gas emissions in transportation processes. As showcased in Figure 8, carbon and resource footprints consistently escalate with a rise in weekly commute frequency, with Scenario 6 exhibiting less favourable results.

Comparing the scenarios against the baseline, an exclusive return to the office results in a 22-ton yearly increase in greenhouse gas emissions, marking a staggering 59% surge. On the contrary, Scenario 1 comes up on the top carbon-saving options, promising a remarkable potential reduction of up to 17% in carbon footprint. Focusing solely on Scenarios 1 and 6 underscores the diminished impacts of working exclusively from home (Scenario 1), showcasing a substantial 48% and 71% reduction in carbon and resource footprints, respectively.

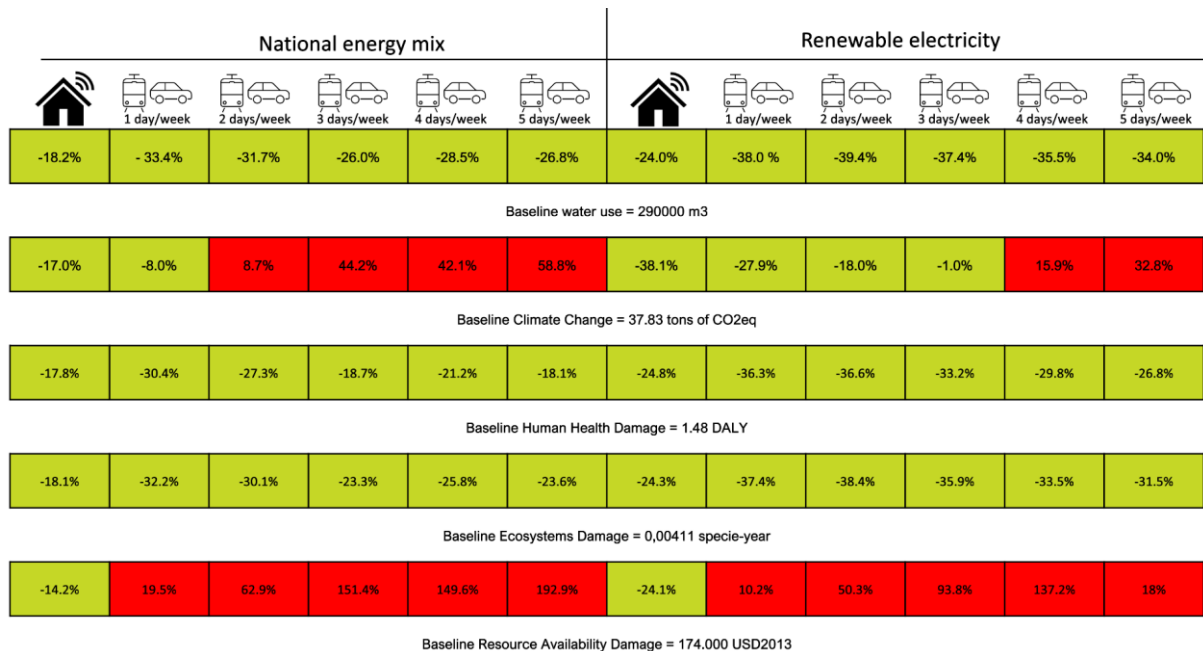
Noteworthy is the fact that among all alternatives, only Scenarios 1 and 2 outperform the baseline in terms of carbon impact. In the realm of resource footprint, scenario 1 stands out as the superior performer. The clear takeaway here is that commuting directly contributes to a negative impact on resource availability. In simple terms, if every staff member commutes two days or more per week, the organization's greenhouse gas emissions will exceed current levels.

Shifting the focus to water consumption and the impacts on the ecosystem and human health, the reduction in Scenario 1 can be attributed to the absence of transport processes. However, in the other scenarios, despite the gradual increase in commuting, the decreased use of the internet for video conferencing and reduced home heating consumption play a role in diminishing these footprints, as both processes are known as water-intensive activities. Notably, scenario 2 emerges as the most effective across all three impact categories.

As depicted in Figure 8, adopting wind-powered electricity further enhances the reduction in all footprints associated with Elfin's production stage.

Figure 8: Production stage-related footprints of compared scenarios.

Including the change in commuting frequency and replacement of electricity source.



5.5 Sensitivity Analysis

Modelling the system involves considering key factors such as methodology, data selection, and assumptions, which may introduce uncertainty to the results.

To ensure the robustness of our findings, a sensitivity analysis (SA) was conducted. As a multi-factor analysis, SA assesses the impact of variations in input parameters on outcomes, gauging the influence of changes in methods, models, values of unmeasured variables, or assumptions. The convergence of primary and sensitivity analysis results enhances the credibility of our findings.

The SA was performed on key parameters to evaluate the significance of our choices. Recognizing the broad applications of SA (Thabane et al, 2013), we strategically focused on parameters with substantial potential to affect global warming impact – energy use, heating, transport (business travel and commuting), and processes linked to the use phase marked by high uncertainty and potential data gaps.

For the first group of processes, characterized by lower uncertainty and significant environmental impact, standard deviations of 10% were assigned. In contrast, the use phase processes (second group) were subjected to higher standard deviations of 30%, acknowledging the higher level of uncertainty.

The sensitivity analysis revealed that climate change impact is highly sensitive to changes in the first group of processes. Even minor alterations (a 10% increase in inputs) could result in a notable 14% rise in the organization's carbon footprint. Conversely, despite a higher deviation assumption, the second group of processes demonstrated minimal impact variation, with a maximum 0.0097% increment in the total climate change impact.

This underscores the reliability of our data, methods, and assumptions, also emphasizing the importance of investing more effort in obtaining precise data for the first group of processes. The Supplementary Information provides detailed insights into the sensitivity analysis.

6. DATA QUALITY ASSESSMENT

A rigorous evaluation of the inventory data employed in this study aims to ascertain its suitability for achieving the study's objectives. Utilizing the pedigree matrix as a tool, foreground processes and data sources underwent assessment across five key dimensions: reliability, completeness, temporal differences, geographical differences, and technological disparities. Table 4 presents a typical pedigree matrix (Weidema et al., 2013), where a score of one (1) signifies the most favorable conditions, while a score of five (5) indicates the least favorable conditions.

Table 4 - Pedigree methodology used to assess data quality.

INDICATOR	SCORE				
	1	2	3	4	5
Reliability	Verified data based on measurements.	Verified data partly based on assumptions or non-verified data based on measurements.	Non-verified data partly based on qualified estimates.	Qualified estimate (e.g., by industrial expert).	Non-qualified estimate.
Completeness	Representative data from all sites relevant to the market considered, over an adequate period to even out normal fluctuations.	Representative data from >50% of the sites relevant for the market considered, over an adequate period to even out normal fluctuations.	Representative data from only some sites (<<50%) relevant for the market considered or >>50% of sites but from shorter periods.	Representative data from only one site relevant for the market considered or some sites but from shorter periods.	Representativeness unknown or incomplete data from a smaller number of sites and from shorter periods.
Temporal records	Less than 3 years of difference to the time-period of the dataset.	Less than 6 years difference to the time period of the dataset.	Less than 10 years difference to the time period of the dataset.	Less than 15 years difference to the time period of the dataset.	Age of data unknown or more than 15 years of difference to the time-period of the dataset.
Geographical records	Data from area under study.	Average data from larger area in which the area under study is included.	Data from area with similar production conditions.	Data from area with slightly similar production conditions.	Data from unknown or distinctly different area.
Technological aspects	Data from enterprises, processes and materials under study.	Data from processes and materials under study but from different enterprises.	Data from processes and materials under study but from different technology	Data on related processes or materials.	Data on related processes on laboratory scale or from different technology.

Table 5 provides the scoring of the datasets with respect to each criterion. Subsequently, the average score was computed. A final score ranging from 1 to 1.6 signifies excellent quality, from 1.6 to 2 indicates very good quality, from 2 to 3 denotes good quality, from 3 to 4 suggests fair quality, and a score exceeding 4 implies poor quality.

Table 5: Inventory data quality assessment.

As indicated in the "Pedigree" column, the numerical sequence represents the scores obtained for reliability, completeness, temporal records, geographical records, and technological aspects, respectively.

STAGE	ACTIVITY	PROCESS	PEDIGREE SCORE	FINAL SCORE
Production	Office and Home-office premises	Home-office electricity consumption	2:3:1:1:2	1.8
		Home-office heating consumption	2:3:1:1:2	1.8
		Office electricity consumption	3:1:1:1:2	1.6
		Office heating consumption	3:1:1:1:2	1.6
		Water consumption	4:1:1:1:2	1.8
	Electronic devices operation	Electricity consumption	1:1:1:1:2	1.2
	Employee commutation	Car/taxi	2:3:1:2:2	2.0
		Train/subway	2:3:1:1:2	1.8
		Bicycle	2:3:1:1:1	1.6
		Walk	2:3:1:1:1	1.6
	Business travel	Car	2:3:1:2:2	2.0
		Train	2:3:1:1:2	1.8
		Bus	2:3:1:1:2	1.8
	Office equipment and furniture	Laptop production	1:1:2:5:2	2.2
		Monitor production	1:1:2:5:2	2.2
		Mouse production	2:1:2:5:2	2.6
		Keyboard Production	1:1:2:5:2	2.2
		Smartphone Production	2:1:2:5:2	2.4
		Office chair Production	3:1:2:5:2	2.6
		Desk Production	3:1:2:5:2	2.6
	Purchased services	Data processing, hosting, and related services	2:1:1:4:1	1.8
		Internet access	2:1:2:3:1	1.8

Use	Use Phase of sold service	Laptop operation	5:5:1:3:2	3.2
		(client) Data hosting service	5:5:1:3:2	3.2
End of Life	Waste generated through office	Waste mix treatment	4:1:1:1:2	1.8
		Wastewater treatment	4:1:1:2:2	2.0
	End of life of office equipment and furniture	Laptop EoL treatment	1:1:2:5:2	2.2
		Monitor production EoL treatment	1:1:2:5:2	2.2
		Mouse production EoL treatment	2:1:2:5:2	2.4
		Keyboard Production EoL treatment	1:1:2:5:2	2.2
		Smartphone Production EoL treatment	2:1:2:5:2	2.6
		Office chair Production EoL treatment	3:1:2:3:2	2.2
		Desk Production EoL treatment	3:1:2:3:2	2.2

Overall, the majority of the production stage processes analyzed received scores ranging from very good to good. Processes with regionalized information within the ecoinvent database achieved the lowest (best) scores, while those relying on global datasets, such as furniture and office equipment production, obtained the highest (worst) scores. This suggests that, in terms of data quality, enhancing regional data granularity was identified as an area for improvement in this study. However, it's crucial to emphasize that this limitation could have been mitigated if manufacturers of capital goods had provided data on their respective productions.

Similar to the production stage, the quality scores for EoL processes presented a comparable pattern. Once again, the lack of regional-specific data contributed to higher process scores, but overall, the scores ranged from very good to good quality.

In contrast, the use phase introduced many uncertainties in defining the organization's final product and how clients would use it. Consequently, processes in this lifecycle stage exhibited the poorest quality.

Despite of the mentioned weaknesses, it is believed that the data and results presented in this study are reliable considering the following factors:

- (i) the background processes found in ecoinvent and associated with the production of capital goods are recent - dating back to 2019. It is expected that these processes have remained consistent over the last three years, with not enough room for significant innovations.
- (ii) given that the value chain for electronic products such as laptops and smartphones is complex and globalized, each component can likely be produced in a different region from where the device is assembled and sold. It is argued, therefore, that using data corresponding to the global average does not deviate much from the local situation.
- (iii) the impacts arising from the use phase are incipient. They do not represent even 0,5% of the organization's water and carbon footprints.
- (iv) when analyzing scenarios, both the baseline and designed scenarios rely on the same datasets, ensuring consistency between assessments.

From this perspective, it is possible to argue that the data, model and results presented here are:

- Precise - since data was calculated or estimated based on site inventory or reliable regional statistical records.
- Comprehensive - as it represents information from all relevant sites. It includes all relevant and traceable upstream, foreground, and downstream processes. All background data comes from ecoinvent's database with documented completeness.
- Consistent - because all primary data were collected with the same level of detail, while all background data were sourced from the ecoinvent database. And all assumptions, methods and data were applied to all scenarios.
- Reproducible - any interested third party can obtain approximate results of this study using the same data and modelling approaches.
- Temporarily representative - All primary data were collected for the year 2022. Secondary data come from recent statistics (2015-2020), and background data was sourced from the most recent version of ecoinvent.
- Geographically representative - whenever possible, preference was given to information about processes occurring within the German territory. Then, in sequence, European Union, neighbouring countries, and finally global data were prioritized. And, in general, most of the selected processes represent a context at least similar to that of the reporting organization.
- Technologically representative - as background data was sourced from the most recent version of ecoinvent.

7. ASSUMPTIONS AND LIMITATIONS

Assumptions are factors that can significantly influence the results of an LCA. Thus, the validity of the results presented in this study is supported by conservative approaches and a clear definition of the limitations of the study in order to avoid potential misinterpretations.

7.1. The process for gathering information on capital goods.

The process of gathering information on capital goods involved the following steps:

1. **Scrutinizing Manufacturers' Websites:**
 - Initially, the manufacturers' websites were scrutinized in search of public Life Cycle Assessment (LCA) reports for specific products or similar items.
 - Most companies publish sustainability reports, and some provide Product Carbon Footprint and Eco Declarations. Unfortunately, in all cases, the required information, including inputs and outputs associated with the production of capital goods, was not found.
2. **Contacting Manufacturers:**
 - The second step involved reaching out to manufacturers. Two attempts were made to obtain the necessary information via email.
 - Only two companies responded. Logitech provided the product's carbon footprint assessment results, while Riitek stated it had not conducted this type of analysis for its portfolio.

Considering the diverse range of capital goods, encompassing various models of laptops, monitors, smartphones, and furniture, and taking into account the limitations imposed by time constraints, as well as the absence of literature presenting the complete Life Cycle Assessment (LCA) inventory or analyzing products with specifications significantly different from those of Elfin, the decision was made to adopt the processes provided by the ecoinvent database as the standard for the analysis of capital goods.

In this case, it was chosen to rely on the processes ("market for") offered by the ecoinvent database as the benchmark for analyzing capital goods. "Market for" processes refer to the consumption mix of a product within a specific region. They encompass inputs from production in one or multiple countries (a collection of country/regional transforming activities) and incorporate inputs from transport processes. Hence, it is deemed that these processes provide a suitable option for representing the diverse range of the company's goods.

Regarding smartphones and office chairs, ecoinvent does not bring any activity representing the production of these material goods. In this way, data from the specialised literature served as the basis for the LCA inventory.

The study by Cordella et al. (2021) was selected for the analysis of smartphones, as it has been conducted recently in the European context and has offered general considerations about a typical smartphone, as opposed to comparing specific devices. Since Elfin has many different models of phones, this last aspect of Cordella's study is something relevant to consider.

Identifying the inputs required to manufacture office chairs was more complex due to the lack of scientific publications that present a comprehensive inventory. It was necessary to call upon the EDP-hub libraries. As EDP-Norge has the highest number of environmental product declarations (EPD) for office furniture, it was the main source of information used. Five similar office chair models targeted at the European market were identified. Using the inventory data from these models, the averages of each required resource were calculated and used in this analysis.

7.2. Extrapolating numbers from the survey and inventory

The survey designed to collect data on home office working spaces received responses from a subset of employees, and average values were subsequently calculated. These averages were considered as a standardized representation for extrapolating data and deriving results that encompassed the entire employee group.

Similarly, for the use phase, a comparable approach was adopted. By analysing the representative file sizes (measured in bytes) and the billing hours expended in creating these documents, extrapolation was possible. This procedure allowed for estimating the number of hours clients spent operating their laptops to review and present projects, as well as quantifying the data stored in clients' hosting services.

7.3. Environmentally extended input-output database

The procurement of services entails impacts extending far upstream, making it impractical to conduct a detailed inventory. Earthster offers a potential solution through the use of the U.S. environmentally extended input-output model (USEEIO). However, it's important to note that USEEIO provides insights solely into the economic context of the United States. Additionally, intrinsic limitations of IO models,

including regional and product aggregation, are present, resulting in values that represent averages. Despite these constraints, USEEIO stands out as the most suitable alternative for evaluating the impacts associated with the purchased service.

7.4. Life Cycle Assessment tools and methods

Certainly, it is essential to emphasize that the outcomes of the life cycle impact assessment are a portrayal of potential environmental impacts rather than actual occurrences. Furthermore, the inventory captures only the environmental load consistent with the chosen approach. Consequently, LCA results should be considered relative expressions and might not accurately reflect the real impact on natural resources and human health.

It is also crucial to note that, despite being a robust, consistent, and comprehensive analysis, inherent limitations exist in the methodological models and assumptions. The data quality and sensitivity analyses indicated the high quality of the presented results. However, these analyses also underscored the importance of obtaining more accurate data, especially about transport and energy processes, in order to enhance the precision and reliability of the results.

8. CONCLUSIONS

This LCA study successfully evaluated the environmental performance of the reporting organization in 2022 and established a framework for ongoing monitoring.

The contribution analysis revealed specific hotspots associated with impact categories of interest, with the production stage being the life cycle phase contributing the most to all impact categories. Processes such as the consumption of electricity, heat, and data processing services systematically dominated the impacts, representing the largest share of responsibility for the company's water and carbon footprints.

It's worth noting that business travel and commuting processes were considerably reduced in 2022 due to pandemic-related restrictions. However, with an escalation in employee attendance at the office, an increase in emissions associated with the transport process and the organization's overall footprint is likely to occur.

The scenario analysis highlighted trade-offs associated with each designed system and choice, providing the organization with alternatives to improve its environmental performance. Options 7 and 8 showed the best results in terms of potential environmental burden reduction. Compared to the baseline, option 8 could minimize the water footprint by up to 38% and the carbon footprint by approximately 28%, while scenario 7 could reduce water consumption by 24% and greenhouse gas emissions by 38%.

Even if the company encourages employees to seek renewable energy providers, changes in the national energy grid are beyond its reach. Therefore, excluding renewable energy from the analysis, scenarios 1 and 2 are identified as the most suitable alternatives for the company's footprint.

As actions to potentially reduce impacts and improve environmental performance, the company may consider:

1. ENCOURAGE SUSTAINABLE COMMUTING:
 - Promote walking, cycling, carpooling, or the use of public transportation among employees to reduce the carbon footprint associated with commuting.
2. IMPLEMENT GREEN PROCUREMENT POLICIES:
 - Establish clear and verifiable environmental criteria for the procurement of products and services, encouraging sustainable practices along the entire supply chain.
3. SELECT ENVIRONMENTALLY FRIENDLY CAPITAL GOODS:
 - Prioritize the purchase of capital goods, such as laptops, monitors, and smartphones, from manufacturers with demonstrated commitment to environmental sustainability, such as Dell, HP, or those following circular business models like FairPhone.
4. OFFSET EMISSIONS:
 - Explore opportunities to offset unavoidable emissions through nature-based solutions (supporting biodiversity restoration and conservation projects) or funding technical carbon removal initiatives (carbon capture, for instance) to achieve carbon neutrality.
5. GREEN ENERGY ADOPTION:
 - Transition to renewable energy sources, such as solar or wind power, to reduce the carbon footprint associated with energy consumption. Encourage employees and partners to embrace sustainability by choosing local renewable energy providers.
6. EMPLOYEE EDUCATION AND ENGAGEMENT:
 - Conduct awareness campaigns and training programs to educate employees about sustainable practices and engage them in environmentally friendly initiatives.
7. WASTE REDUCTION AND RECYCLING PROGRAMS:
 - Implement effective waste reduction strategies, including recycling programs and waste audits, to minimize the environmental impact of waste generated by the company.
8. ENERGY EFFICIENCY MEASURES:
 - Invest in energy-efficient technologies and practices to reduce overall energy consumption and decrease the environmental impact associated with energy production.
9. CARBON REPORTING AND TRANSPARENCY:
 - Regularly report and communicate the company's carbon footprint and environmental performance to stakeholders, fostering transparency and accountability.
10. CONTINUOUS MONITORING AND IMPROVEMENT:
 - Establish a robust monitoring and reporting system to regularly assess environmental performance, identify areas for improvement, and implement ongoing sustainability initiatives.

By incorporating these actions into its operational strategies, Elfin can make a positive impact on the environment while supporting local green initiatives, contributing to a more sustainable and environmentally responsible business model.

Finally, to stay up to date and get more accurate scenarios, the reporting organization must strive to collect more accurate data on energy consumption and transport processes (commuting and business travel).

9. REFERENCE:

IPCC - Intergovernmental Panel on Climate Change. 2022. Summary for Policymakers.

In: Climate Change 2022: Mitigation of Climate Change. Contribution of Working Group III to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change [P.R. Shukla, J. Skea, R. Slade, A. Al Khourdajie, R. van Diemen, D. McCollum, M. Pathak, S. Some, P. Vyas, R. Fradera, M. Belkacemi, A. Hasija, G. Lisboa, S. Luz, J. Malley, (eds.)]. Cambridge University Press, Cambridge, UK and New York, NY, USA. doi: 10.1017/9781009157926.001.

ISO 14040, 2006. ISO 14040: Environmental management – Life cycle assessment – Principles and framework. Geneva: International Organization for Standardization.

ISO 14044, 2006. ISO 14044: Environmental management – Life cycle assessment – Requirements and guidelines. Geneva: International Organization for Standardization.

ISO. ISO/TS 14072: Environmental Management—Life Cycle Assessment—Requirements and Guidelines for Organizational Life Cycle Assessment; International Organization for Standardization: Geneva, Switzerland, 2014.

Mytton, D. Data centre water consumption. npj Clean Water 4, 11 (2021).

<https://doi.org/10.1038/s41545-021-00101-w>

Ou et al., 2015 - Life cycle water use of coal- and natural-gas-fired power plants with and without carbon capture and storage. International Journal of Greenhouse Gas Control, Volume 44, January 2016, Pages 249-261. <https://doi.org/10.1016/j.ijggc.2015.11.029>

UNEP. Guidance on Organizational Life Cycle Assessment; UNEP: Paris, France, 2015; ISBN 978-92-807-3453-9.

Weidema, B.P., Bauer, C., Hischier, R., Mutel, C., Nemecek, T., Reinhard, J., Vadenbo, C.O., Wernet, G. 2013. Overview and methodology. Data quality guideline for the ecoinvent database version 3. Ecoinvent Report 1(v3). The ecoinvent Centre. St. Gallen, Switzerland

Weidema, B.P., Wesnaes, M.S. 1996. Data quality management for life cycle inventories – an example of using data quality indicators. J Clean Prod 4(3-4): 167-174